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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Loan Eligibility Verification and EMI Payment Tracking System

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DECLARATION

I hereby declare that the project entitled "**Loan Eligibility Verification and EMI Payment Tracking System"** submitted in partial fulfillment of the requirements for the course [**Java Programming]** is a **record of my original work** carried out under the guidance of **[Dharaneesh.s]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Deva.G

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Loan Eligibility Verification and EMI Payment Tracking System"** is a **bona fide work** carried out by **[Akash.S] [B.Sc(CS)-1C]**, [**COMPUTER SCIENCE]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Java Programming]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
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ABSTRACT

The Loan Eligibility Verification and EMI Payment Tracking System is a Java-based application designed to automate and simplify the process of evaluating loan eligibility and managing EMI (Equated Monthly Installment) payments. In traditional banking systems, loan approval and EMI tracking involve manual calculations, paperwork, and time-consuming verification processes. These methods are prone to human error, inefficiency, and delays.

This system provides a structured and automated approach using core Java programming concepts. It allows users to input customer details such as income, credit score, and loan requirements, and then determines loan eligibility based on predefined criteria. Additionally, the system calculates EMI values and tracks payment schedules efficiently.

The application is developed as a console-based system to maintain simplicity and focus on logic implementation. It ensures accuracy in financial calculations, reduces manual workload, and improves decision-making. This project demonstrates the practical application of object-oriented programming concepts in solving real-world financial problems.

1. INTRODUCTION

In today's financial environment, banks and financial institutions handle a large number of loan applications daily. Evaluating each application manually can be time-consuming and inefficient. It requires analyzing customer income, credit score, and repayment capacity before approving a loan.

Loan management also involves tracking EMI payments over a period of time. Manual tracking can lead to missed payments, incorrect calculations, and poor record management.

To overcome these challenges, the Loan Eligibility Verification and EMI Payment Tracking System is developed. This system automates the process of verifying loan eligibility and calculating EMI values. It also maintains a structured record of payments, ensuring transparency and accuracy.

The system is implemented using Java programming language and follows object-oriented design principles. It provides a reliable and efficient solution for managing loan-related operations.

1.1 Objective

The primary objectives of this project are:

- To design and develop a system for loan eligibility verification
- To automate EMI calculation based on loan parameters
- To reduce manual errors in financial calculations
- To track EMI payments efficiently
- To implement object-oriented programming concepts
- To provide a simple and user-friendly interface
- To improve accuracy and speed in loan processing

1.2 Overview

This system is a menu-driven Java application that performs multiple operations related to loan processing. It allows users to input customer details and loan requirements, after which it evaluates eligibility and calculates EMI.

The system uses structured classes to manage customer and loan data. It ensures modularity, making the application easy to understand and maintain.

Key operations include:

- Customer data entry
- Loan eligibility checking
- EMI calculation
- Payment tracking
- Report generation

1.3 Features

The system offers the following features:

- Easy input of customer details
- Automatic eligibility verification
- Accurate EMI calculation
- Payment tracking system
- Structured output display
- Efficient handling of multiple customers
- Fast processing and reliable results

2. PROGRAM ANALYSIS

2.1 Problem Definition

Traditional loan processing systems face several challenges:

- Manual calculations leading to errors
- Time-consuming verification process
- Difficulty in managing multiple customers
- Lack of proper EMI tracking
- Inefficient record maintenance

These issues highlight the need for an automated system that can handle loan processing efficiently.

2.2 Project Overview

The project provides a complete solution to loan management problems by introducing an automated system. It simplifies the process of determining loan eligibility and calculating EMI values.

The system performs the following operations:

- Stores customer details
- Verifies eligibility based on criteria
- Calculates EMI using formulas
- Tracks payment details
- Displays reports

2.3 Scopes and Limitations

Scope:

- Banks and financial institutions
- Small lending organizations
- Educational purposes

Limitations:

- Console-based interface (no GUI)
- No database integration
- Limited to basic functionality

3. SYSTEM DESIGN

The system is divided into multiple modules to ensure proper organization and efficient functioning.

3.1 Modules

1. Customer Module

This module handles all customer-related data. It stores details such as:

- Name
- Monthly income
- Credit score

It ensures that customer data is organized and easily accessible.

2. Loan Module

This module manages loan-related information, including:

- Loan amount
- Interest rate
- Loan tenure

It acts as the core component for financial calculations.

3. Eligibility Module

This module determines whether a customer is eligible for a loan. The eligibility criteria may include:

- Minimum income requirement
- Credit score threshold

The system evaluates these conditions and provides a decision.

4. EMI Calculation Module

This module calculates EMI using a standard formula. It ensures accurate financial computation and helps customers understand their repayment obligations.

5. Payment Tracking Module

This module keeps track of EMI payments. It records:

- Paid installments
- Remaining balance
- Payment schedule

It ensures proper monitoring of loan repayment.

6. Report Module

This module generates detailed output, including:

- Customer details
- Loan information
- EMI values
- Payment status

4. METHODOLOGY

The development of this system follows a structured methodology to ensure accuracy and efficiency.

4.1 PROBLEM ANALYSIS

The first step involves studying the existing system and identifying its drawbacks. Manual loan processing is inefficient and error-prone, which necessitates automation.

4.2 SYSTEM DESIGN

The system is divided into modules to simplify development. Each module performs a specific task, ensuring clarity and maintainability.

4.3 PROGRAM DEVELOPMENT

The system is implemented using Java with object-oriented programming concepts such as:

- Classes and Objects
- Methods
- Conditional statements
- Loops

The code is structured to ensure readability and reusability.

4.4 TESTING AND VALIDATION

The system is tested using various inputs to ensure correctness. Test cases include:

- Different income levels
- Various loan amounts
- Edge cases

Testing ensures accurate results and error-free execution.

4.5 RESULT AND DOCUMENTATION

The final stage includes documenting the system, explaining its functionality, and presenting results.

4.6 ALGORITHM

1. Start the program
2. Enter customer details
3. Input loan details
4. Check eligibility
5. If eligible, calculate EMI
6. Track payment details
7. Display results
8. End

4.7 FLOWCHART (TEXT REPRESENTATION)

Start

↓

Enter Customer Details

↓

Check Eligibility

↓

Eligible? → Yes → Calculate EMI

↓

Track Payments

↓

Display Output

↓

End

5. SOURCE CODE EXPLANATION

The Loan Eligibility Verification and EMI Payment Tracking System is developed using a modular and object-oriented approach. The system is divided into multiple classes, where each class is responsible for a specific functionality. This design improves code readability, maintainability, and scalability.

1. Customer Class

The **Customer Class** is responsible for storing and managing all customer-related information. It acts as the foundation of the system because loan eligibility decisions are based on customer data.

Attributes:

- **Name** – Stores the name of the customer
- **Monthly Income** – Represents the earning capacity of the customer
- **Credit Score** – Indicates the financial reliability of the customer

Functions:

- Accepts customer details as input
- Provides methods to retrieve customer information
- Validates basic input values (e.g., income should be positive)

Importance:

This class plays a crucial role in determining whether a customer qualifies for a loan. A higher income and better credit score increase the chances of eligibility.

2. Loan Class

The **Loan Class** manages all loan-related operations. It stores loan details and performs financial calculations required for EMI determination.

Attributes:

- **Loan Amount** – Total amount requested by the customer
- **Interest Rate** – Annual interest rate applied to the loan
- **Tenure** – Duration of the loan in months
- **EMI** – Monthly installment amount to be paid

Functions:

- Calculates EMI using a standard formula
- Stores and updates loan-related values
- Provides methods to display loan details

EMI Calculation Formula:

EMI=P · R · (1+R)N(1+R)N−1EMI = \frac{P \cdot R \cdot (1+R)^N}{(1+R)^N - 1}EMI=(1+R)N−1P · R · (1+R)N

Where:

- **P** = Loan Amount
- **R** = Monthly Interest Rate (Annual Rate / 12 / 100)
- **N** = Number of months

Importance:

This class ensures accurate financial calculations and helps users understand their repayment commitments clearly.

3. Main Class

The **Main Class** acts as the controller of the entire application. It manages user interaction and coordinates between different classes.

Responsibilities:

- Displays menu options to the user
- Accepts input from the user
- Calls methods from Customer and Loan classes
- Controls the flow of the program

Typical Workflow:

1. User enters customer details
2. System checks loan eligibility
3. If eligible, loan details are entered
4. EMI is calculated
5. Results are displayed

Concepts Used

- Object-Oriented Programming
- Conditional Logic
- Loops
- Modular Programming

6.INPUT AND OUTPUT SCREENS

Input

- Customer name
- Monthly income
- Credit score
- Loan amount
- Interest rate
- Tenure

Output

- Eligibility status
- EMI amount
- Payment details
- Remaining balance

7.DISCUSSION & CONCLUSION

7. RESULTS AND DISCUSSION

The system successfully performs all required operations. It accurately determines loan eligibility and calculates EMI values. It also tracks payments effectively.

The results show that automation significantly improves efficiency and reduces errors compared to manual systems.

7.1 DISCUSSION

This project demonstrates how programming can solve real-world financial problems. The system reduces workload and enhances accuracy.

The modular design ensures scalability and easy maintenance.

7.2 CONCLUSION

The Loan Eligibility Verification and EMI Payment Tracking System is a successful implementation of a financial application using Java. It simplifies loan processing and ensures accurate calculations.

The project meets all objectives and demonstrates practical use of programming concepts.

FUTURE ENHANCEMENTS

The system can be improved by adding:

- Graphical User Interface (GUI)
- Database integration (MySQL)
- Web-based interface
- Mobile application
- Advanced analytics and reports

8.REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com